

Q/XNP 40-2024

Technical Specification for the Application of Self-Fusing Functional Film for Architectural Metal Surface Protection

Company Standard of Xiniupi Waterproof Technology Co., Ltd.

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Translator's note: This English version translates the standards text, figure captions, and figure callout legends from the source PDF. The original line drawings are not redrawn in this text version; use the translated captions and callout legends together with the original figures where needed.

Foreword

This specification was drafted in accordance with the rules given in GB/T 1.1-2020.

The drafting group prepared this specification after extensive investigation and research, careful summary of practical experience, reference to relevant advanced international and domestic standards and practices, and broad consultation.

This specification consists of 8 chapters. The main technical contents include: general provisions, terms, basic requirements, materials, design, construction, acceptance, and operation and maintenance.

This specification was proposed, drafted, and is interpreted by Xiniupi Waterproof Technology Co., Ltd. It applies to the design, construction, acceptance, operation, and maintenance of metal surface protection for existing and new buildings and structures. If amendments or additions are needed during use, please provide feedback to Xiniupi Waterproof Technology Co., Ltd. Address: Xiniupi Science and Technology Park Research Building, No. 68, Team 4, Naluopo, Sitang Community, Santang Town, Xingning District, Nanning, Guangxi Zhuang Autonomous Region, China. Postal code: 530216. Tel: 0771-5623151. Email: xnp@xnpfs.com. Feedback will be used as reference for future revisions.

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Contents

1. General Provisions
2. Terms
3. Basic Requirements
4. Materials
5. Design
 - 5.1 General Requirements
 - 5.2 Basic Structural Arrangement
 - 5.3 Detail Design for Metal Roof Protection

- 5.4 Detail Design for Metal Wall Protection

6. Construction

- 6.1 General Requirements

- 6.2 Metal Roof Protection Construction

- 6.3 Metal Wall Protection Construction

- 6.4 Green Construction

7. Acceptance

- 7.1 General Requirements

- 7.2 Acceptance of Metal Roof Protection

- 7.3 Acceptance of Metal Wall Protection

8. Operation and Maintenance

Additional sections:

- Explanation of Wording

- List of Referenced Standards

- Addition: Explanation of Provisions

1 General Provisions

1.0.1

This specification is formulated to standardize the application of self-fusing functional film for architectural metal surface protection, achieve advanced technology, safety and durability, energy saving and environmental protection, and economic rationality, and ensure project quality.

1.0.2

This specification applies to the design, construction, acceptance, operation, and maintenance of protection works using self-fusing functional film for architectural metal surface protection on existing and new buildings and structures.

1.0.3

In addition to complying with this specification, the application of self-fusing functional film for architectural metal surface protection shall also comply with the provisions of current relevant national standards.

2 Terms

2.0.1 Self-fusing functional film for architectural metal surface protection (hereinafter referred to as functional protective film)

A functional film composed of a weather-resistant polymer film facing material, an adhesive layer, and a release protective film. It can fuse and bond with the surfaces of metal buildings and structures, can be used exposed, and provides protective functions such as leakage prevention and corrosion prevention.

2.0.2 Fusion dry-bonding method

A construction method in which the functional protective film is directly adhered to the substrate surface after removal of the release protective layer, using the material's own fusion-bonding characteristics to achieve tight adhesion to the substrate.

2.0.3 Adhesive bonding method

A construction method in which the functional protective film is adhered to a cement-based substrate surface using cement-based adhesive material, or another adhesive material with good bonding performance and compatibility, as the bonding material.

3 Basic Requirements

3.0.1

Projects using functional protective film shall be designed according to the importance of the building or structure, regional climate, and durability requirements.

3.0.2

When functional protective film is installed on a metal surface, the fusion dry-bonding method should be used. When it is installed on the surface of cement-based materials, the adhesive bonding method should be used. When it is installed on substrate surfaces such as stone, wood, or plastic, construction may refer to the practice for metal substrate surfaces, but trial adhesion shall be performed on site before construction, and large-area construction may proceed only after the bonding effect is confirmed to meet the application technical requirements.

3.0.3

Where a metal structural part no longer has the required service function of the metal structure due to corrosion, it shall be replaced.

3.0.4

Other structural designs shall refer to this specification and current relevant standards.

4 Materials

4.0.1

The functional protective film shall comply with the relevant requirements of the company standard Functional Film for Architectural Metal Surface Protection, Q/XNP 34-2024. Its main physical properties shall comply with Table 4.0.1.

Table 4.0.1 Main Physical Properties of Functional Protective Film

No.	Item	Technical Requirement
1	Tensile properties - tensile force, N/50 mm	≥ 450
1	Tensile properties - elongation at maximum tensile force, %	≥ 50
1	Tensile properties - condition during tensile test	No separation between adhesive layer and facing material
2	Tear force, N	≥ 20
3	Peel strength, N/mm, to functional film overlap edge, untreated	≥ 1.0
3	Peel strength, N/mm, to metal, untreated	≥ 1.0
4	Heat resistance, 100 deg C, 2 h	No flowing or dripping; sliding ≤ 2 mm
5	Low-temperature flexibility, -40 deg C	No cracks
6	Thermal-oxidative aging, 80 deg C, 7 d, tensile force retention, %	≥ 90
6	Thermal-oxidative aging, 80 deg C, 7 d, elongation retention, %	≥ 80

6	Thermal-oxidative aging, 80 deg C, 14 d, low-temperature flexibility	No cracks at -38 deg C
7	Artificial accelerated weathering, ≥ 2745 h	No blistering, cracks, or delamination; no adhesion of facing material; no holes
8	Watertightness, 0.3 MPa, 120 min	Watertight

4.0.2

When the adhesive bonding method is used, the cement used to prepare the cement-based adhesive material shall be Portland cement or ordinary Portland cement with a strength grade not lower than 42.5, and the water-cement ratio should not be greater than 0.45.

5 Design

5.1 General Requirements

5.1.1

The design of functional protective film shall follow the principles of adapting to local conditions, preventing leakage and corrosion, combining prevention and drainage, and ensuring structural safety.

5.1.2

The minimum overlap width of functional protective film shall not be less than 30 mm.

5.1.3

For detail nodes, whether an additional layer is provided shall be determined according to the actual project conditions. Where an additional layer is provided, it may be arranged with reference to the following practices:

1. The additional layer material shall be functional protective film or a compatible sealing material.
2. Additional layers at fixing-point nail holes and joints shall be reinforced with matching node sealing tape or compatible sealing material. Deformation allowance shall be reserved in the additional layer, and the distance from the edge of the additional layer to the edge of the joint should not be less than 30 mm on each side.

5.2 Basic Structural Arrangement

5.2.1

Functional protective film shall be installed directly on the surfaces of metal buildings and structures.

5.2.2

The basic structural arrangement of functional protective film is shown in Figure 5.2.2.

Legend: 1 - functional protective film; 2 - metal substrate surface.

Figure 5.2.2 Basic structural arrangement.

5.2.2

The metal substrate surface includes metal tiles and metal panels with surface coating. Where coating chalking, delamination, or similar conditions affect the bonding between the functional protective film and the substrate surface, the surface shall be cleaned before the functional protective film is installed.

5.3 Detail Design for Metal Roof Protection

5.3.1

The protective construction for a double-slope metal roof ridge is shown in Figure 5.3.1-1, and the protective construction for a single-slope roof ridge is shown in Figure 5.3.1-2.

Legend for Figure 5.3.1-1: 1 - functional protective film; 2 - ridge cover plate; 3 - water stop plate; 4 - backing purlin; 5 - purlin.

Figure 5.3.1-1 Protective construction for double-slope roof ridge.

Legend for Figure 5.3.1-2: 1 - ridge edge-trim support; 2 - ridge panel; 3 - functional protective film; 4 - backing purlin; 5 - purlin.

Figure 5.3.1-2 Protective construction for single-slope roof ridge.

5.3.2

The protective construction for roof eaves is shown in Figure 5.3.2.

Legend: 1 - functional protective film; 2 - eaves closure; 3 - backing purlin; 4 - eaves edge trim.

Figure 5.3.2 Protective construction for roof eaves.

5.3.3

The protective construction design for roof eaves gutters and valley gutters shall comply with the following provisions:

1. The functional protective film in eaves gutters and valley gutters shall be overlapped in the direction of water flow. It shall turn up from the gutter bottom to the top of the outer side, and the functional protective film should be adhered into the rainwater outlet, as shown in Figures 5.3.3-1 to 5.3.3-2.
2. When the eaves gutter or valley gutter is a concrete or masonry plastered surface, the functional protective film should be installed using the adhesive bonding method, turning up from the gutter bottom to the top of the outer side.

Legend for Figure 5.3.3-1: 1 - rainwater pipe; 2 - metal eaves gutter; 3 - eaves edge trim; 4 - eaves gutter tie rod; 5 - eaves closure; 6 - functional protective film; 7 - backing purlin.

Figure 5.3.3-1 Protective construction for eaves gutter.

Legend for Figure 5.3.3-2: 1 - functional protective film; 2 - tie rod; 3 - metal valley gutter; 4 - rainwater pipe.

Figure 5.3.3-2 Protective construction for valley gutter.

5.3.4

The protective construction for a low parapet is shown in Figure 5.3.4-1, and the protective construction for a high parapet is shown in Figure 5.3.4-2.

Legend for Figure 5.3.4-1: 1 - functional protective film; 2 - edge trim; 3 - backing purlin; 4 - backing purlin support.

Figure 5.3.4-1 Protective construction for low parapet.

Legend for Figure 5.3.4-2: 1 - functional protective film; 2 - termination seal; 3 - edge trim; 4 - backing purlin; 5 - backing purlin support.

Figure 5.3.4-2 Protective construction for high parapet.

5.3.5

The protective construction for a roof expansion joint is shown in Figure 5.3.5.

Legend: 1 - backing material; 2 - expansion-joint cover plate; 3 - functional protective film; 4 - backing purlin; 5 - roof purlin.

Figure 5.3.5 Protective construction for roof expansion joint.

5.3.6

The protective construction for a pipe penetrating the roof is shown in Figure 5.3.6. The protective height of the functional protective film along the pipe root should not be less than 250 mm.

Legend: 1 - pipe penetrating the roof; 2 - termination seal; 3 - functional protective film; 4 - roof purlin.

Figure 5.3.6 Protective construction for pipe penetrating the roof.

5.3.7

The protective construction for photovoltaic fixing parts is shown in Figure 5.3.7. The functional protective film should be cut off at the fixing position between the adapter and the support, and the cut-off area shall be sealed tightly with compatible sealing material. Where photovoltaic roof supports pass through the metal roof, damaged areas of the functional protective film shall be sealed.

Legend: 1 - aluminum alloy connector; 2 - photovoltaic component; 3 - aluminum alloy adapter; 4 - termination seal; 5 - T-shaped support; 6 - purlin; 7 - functional protective film; 8 - metal roof.

Figure 5.3.7 Protective construction for photovoltaic fixing parts.

5.3.8

The protective construction for a ventilator is shown in Figure 5.3.8. The functional protective film shall be laid up to the ventilator neck, and the termination shall be sealed tightly with compatible sealing material.

Legend: 1 - roof purlin; 2 - flashing plate; 3 - functional protective film; 4 - ventilation ball.

Figure 5.3.8 Protective construction for ventilator.

5.4 Detail Design for Metal Wall Protection

5.4.1

The protective construction for a wall expansion joint is shown in Figure 5.4.1.

Legend: 1 - metal wall panel; 2 - expansion-joint cover plate; 3 - functional protective film.

Figure 5.4.1 Protective construction for wall expansion joint.

5.4.2

The protective construction for the window side is shown in Figure 5.4.2-1, and the protective construction for the window head and window sill is shown in Figure 5.4.2-2. The functional protective film shall be laid to the edge of the window frame, and the termination shall be sealed tightly with compatible sealing material.

Legend for Figure 5.4.2-1: 1 - metal wall panel; 2 - functional protective film; 3 - external corner trim at window side; 4 - termination seal.

Figure 5.4.2-1 Protective construction for window side.

Legend for Figure 5.4.2-2: 1 - metal wall panel; 2 - functional protective film; 3 - external corner trim at window head; 4 - termination seal; 5 - external corner trim at window sill.

Figure 5.4.2-2 Protective construction for window head and window sill.

6 Construction

6.1 General Requirements

6.1.1

Functional protective film shall be installed by a professional team, and operators shall take up their posts only after passing professional training.

6.1.2

Before construction of functional protective film, site investigation and drawing review shall be carried out. Operating conditions and the coordination and implementation of safety measures shall be discussed with relevant parties. A special construction plan or technical measures shall be prepared and reviewed and confirmed by relevant parties, and on-site technical and safety disclosure shall then be conducted before construction.

6.1.3

The functional protective film shall have product certificates and factory performance test reports. The material variety, specifications, performance, and other properties shall comply with the requirements of the design or special plan and product standards.

6.1.4

The storage, transportation, and custody of functional protective film shall comply with the following provisions:

1. Products of different types and specifications shall be stacked separately and shall not be mixed.
2. Sun exposure and rain shall be avoided; ventilation shall be maintained; and products shall be kept away from ignition sources.
3. The storage temperature should be -5 deg C to 45 deg C.
4. Functional protective film should be stored upright in a single layer. When stored flat, the stacking height shall not exceed 5 layers.

6.1.5

Construction tools and equipment shall be prepared according to construction needs. Common construction tools and equipment mainly include:

1. Substrate treatment tools: blower, broom, roller, brush, grinder, hand drill, etc.
2. Laying tools: scraper, utility knife, caulking gun, steel tape measure, etc.
3. Safety protection equipment: safety helmet, anti-slip shoes, gloves, safety rope, safety belt, hook, etc.

6.1.6

Before construction of functional protective film, substrate surface quality shall comply with the following provisions:

1. Reinforced or replaced parts of the metal substrate surface have passed acceptance.
2. Rusted parts have been derusted or treated with rust stabilizer and have passed acceptance.
3. The metal substrate surface is dry and free of stains, dust, and other loose matter.

6.1.7

Facilities, pipes, embedded parts, and similar items penetrating the metal surface shall be installed and fixed before construction.

6.1.8

Before construction, safety inspection shall be carried out on the machinery, protective equipment, and surrounding operating environment to be used.

6.1.9

Outdoor functional protective film construction shall not be carried out in rainy weather, snowy weather, or wind of Grade 5 or above.

6.1.10

The construction ambient temperature for functional protective film should be 5 deg C to 45 deg C. When the construction temperature is too low, heating measures to assist adhesion should be adopted.

6.1.11

Construction safety for functional protective film shall comply with the following requirements:

1. Where there are no reliable protective facilities around the working surface or at reserved openings, fall-prevention measures shall be adopted.
2. When laying roof functional protective film, temporary pedestrian walkway panels and material access routes should be provided on the metal panel surface. Roof materials shall be dispersed to reduce safety impacts on structural load-bearing.
3. Waste generated during construction shall be collected promptly to prevent falling and shall not be discarded arbitrarily.

6.1.12

Overlap joints of functional protective film shall comply with the following provisions:

1. The overlap joint width shall comply with Article 5.1.2 of this specification.
2. Short-side overlap joints of two adjacent sheets of functional protective film in the same layer shall be staggered by not less than 300 mm.
3. Long-side overlap joints of upper and lower layers of functional protective film shall be staggered, and the staggered distance shall not be less than one-quarter of the sheet width.
4. During functional protective film construction, overlaps should be arranged so as to avoid panel joints.

6.1.13

When multiple processes are involved on the metal surface of a building or structure, the functional protective film should be installed before other processes.

6.1.14

After each process is completed, inspection and acceptance shall be carried out and complete inspection records shall be kept. The next process may proceed only after acceptance. During construction of adjacent processes, completed parts shall be cleaned and protected. If the functional protective film is damaged, it shall be repaired promptly.

6.2 Metal Roof Protection Construction

6.2.1

The laying sequence and direction of functional protective film shall comply with the following provisions:

1. Trial laying and positioning shall be carried out before laying.
2. The film should be laid perpendicular to the crest of the metal panel ribs. When laid in two layers, the upper and lower layers of functional protective film should not be laid perpendicular to each other.
3. Construction should proceed upward from the lowest elevation of the roof. When laying roofs with multiple spans or high and low spans, construction shall proceed in the order of higher before lower and farther before nearer.
4. When functional protective film is installed in eaves gutters and valley gutters, overlap joints should be in the direction of water flow.

6.2.2

Functional protective film around rainwater outlets and drain hoppers should be turned into the hopper and adhered tightly.

6.3 Metal Wall Protection Construction

6.3.1

Construction on external metal walls should proceed in the sequence of detail node treatment first, followed by large-area construction.

6.3.2

When functional protective film is laid on external metal walls, it should be laid from low to high.

6.3.3

The termination of functional protective film at window openings or other openings shall be fixed on the inner side of the opening. Connections with door and window frames shall be sealed tightly with compatible sealing material.

6.4 Green Construction

6.4.1

Rust stabilizers, rust removers, and sealing materials shall be environmentally friendly materials.

6.4.2

Release protective materials and paper tube cores removed from functional protective film shall be recycled promptly.

6.4.3

Appropriate protective equipment shall be provided during construction of functional protective film.

7 Acceptance

7.1 General Requirements

7.1.1

The following technical documents shall be submitted at project acceptance:

1. Product certificates, product performance test reports, and on-site sampling inspection reports for incoming materials.
2. Construction plan.
3. Other documents that shall be submitted.

7.1.2

After functional protective film enters the construction site, sampling inspection shall be carried out as required. Products of the same type and specification shall be grouped into one batch for each 10000 m², and quantities less than 10000 m² shall also be counted as one batch. Five rolls shall be randomly selected from each batch for thickness, area, and appearance inspection. After the above inspections are passed, one roll shall be randomly selected and a sample of not less than 1.5 m² shall be taken for physical property testing.

7.1.3

Incoming inspection items for functional protective film include appearance, area, thickness, tensile properties, tear force, heat resistance, low-temperature flexibility, and peel strength between functional film and functional film (untreated).

7.1.4

After construction of functional protective film on a metal roof is completed, post-rain observation or water spraying or water storage tests shall be carried out. Where water storage testing is not feasible due to structural load-bearing or other reasons, the water storage test may be omitted, but the water spraying test time shall be extended and the following provisions shall be met:

1. When post-rain observation is used, the rainfall shall reach the moderate-rainfall grade.
2. When a water spraying test is used, the continuous spraying time shall not be less than 2 h.
3. Eaves gutters, valley gutters, rainwater outlets, and similar parts shall be subjected to water storage testing. The minimum water storage height shall not be less than 20 mm, and the storage time shall not be less than 24 h.

7.1.5

After construction of functional film for architectural metal exterior wall surface protection is completed, a water spraying test shall be carried out and shall comply with the following provisions:

1. The continuous water spraying time shall not be less than 30 min.
2. For building exterior walls where only nodes such as doors and windows are protected, water spraying tests may be carried out only on nodes such as doors and windows.

7.1.6

The random inspection quantity for inspection lots of functional protective film construction sub-items shall be one location for each 100 m² of laid area. Each location shall be 10 m², and there shall be not less than 3 locations.

7.2 Acceptance of Metal Roof Protection

I Main Control Items

7.2.1

The functional protective film shall comply with the design requirements.

Inspection method: check factory certificate, quality inspection report, and incoming inspection report.

7.2.2

There shall be no leakage after construction of the functional protective film is completed.

Inspection method: post-rain observation or water spraying or water storage test.

7.2.3

The detail construction of functional protective film at valley gutters, eaves gutters, eaves, and similar parts shall comply with the design provisions.

Inspection method: visual inspection and checking concealed work acceptance records.

II General Items

7.2.4

The substrate surface shall be clean, dry, neat, and firm, free of attachments such as dust, rust, and frozen matter, free of oil or sundries, and free of visible water.

Inspection method: visual inspection and checking concealed work acceptance records.

7.2.5

The functional protective film and metal substrate surface shall be firmly bonded.

Inspection method: visual inspection.

7.2.6

The laying direction of functional protective film shall comply with the design requirements.

Inspection method: visual inspection.

7.2.7

The minimum overlap width of functional protective film shall not be less than 30 mm.

Inspection method: measurement with ruler.

7.3 Acceptance of Metal Wall Protection

I Main Control Items

7.3.1

The quality of the functional protective film and its matching materials shall comply with the design requirements.

Inspection method: check factory certificate, quality inspection report, and incoming inspection report.

7.3.2

There shall be no leakage after construction of the functional protective film is completed.

Inspection method: observe and inspect after continuous water spraying for 30 min.

7.3.3

The node practices of functional protective film at door and window openings, pipes extending through exterior walls, embedded parts, and similar parts shall comply with the design requirements.

Inspection method: visual inspection and checking concealed work acceptance records.

II General Items

7.3.4

The functional protective film and metal substrate surface shall be firmly bonded.

Inspection method: visual inspection.

7.3.5

The minimum overlap width of functional protective film shall not be less than 30 mm.

Inspection method: measurement with ruler.

8 Operation and Maintenance

8.0.1

A maintenance management system for the protection works shall be established, and regular inspection and maintenance shall be carried out.

8.0.2

When the function of the functional protective film fails, maintenance or renovation shall be carried out immediately.

8.0.3

An emergency leakage response plan shall be established for the protection works.

8.0.1

Materials and processes used for repair of the protection works shall not produce harmful physical or chemical effects.

8.0.2

Safety and quality assurance measures shall be adopted for on-site maintenance or repair operations. Outdoor high-altitude operations shall not be carried out when gusts are Grade 5 or above.

Explanation of Wording

For ease of distinguishing the degree of strictness of requirements when implementing the provisions of this specification, wording is explained as follows:

1. Words indicating a very strict requirement that must be followed: the positive word is "must"; the negative word is "strictly prohibited".
2. Words indicating a strict requirement that shall normally be followed: the positive word is "shall"; the negative words are "shall not" or "must not".
3. Words indicating that a slight choice is allowed and that the stated approach should be adopted first where conditions permit: the positive word is "should"; the negative words are "should not".
4. Words indicating that a choice is allowed and that the approach may be adopted under certain conditions: "may" is used.

List of Referenced Standards

This specification references the following standard. For dated references, only the version corresponding to that date applies to this specification; for undated references, the latest version applies to this specification.

- Functional Film for Architectural Metal Surface Protection, Q/XNP 34-2024

Addition: Explanation of Provisions

Technical Specification for the Application of Self-Fusing Functional Film for Architectural Metal Surface Protection - Explanation of Provisions

Drafting Explanation

During preparation of the Technical Specification for the Application of Self-Fusing Functional Film for Architectural Metal Surface Protection, the drafting group conducted extensive investigation and research, summarized practical experience in protective engineering construction in China, referred to advanced foreign technical regulations and technical standards, and obtained many important technical parameters through testing.

The drafting group prepared the explanation of provisions for this specification in the order of chapters, sections, and articles. The explanation describes the purpose and basis of the provisions and matters requiring attention during implementation. This explanation of provisions does not have the same legal effect as the main text and appendices of the standard, and is provided only as a reference for users to understand and apply the standard provisions.

Contents of Explanation of Provisions

1. General Provisions
2. Terms
3. Basic Requirements
4. Materials
5. Design
 - 5.1 General Requirements
 - 5.3 Detail Design for Metal Roof Protection
6. Construction
 - 6.1 General Requirements
 - 6.2 Metal Roof Protection Construction
 - 6.3 Metal Wall Protection Construction
7. Acceptance
 - 7.1 General Requirements
8. Operation and Maintenance

1 General Provisions

1.0.2

Functional protective film applies to the protection of metal substrate surfaces of existing and new buildings and structures. Types of metal buildings and structures include residences, factories, office buildings, shopping centers, schools, stations, airports, gymnasiums, warehouses, oil storage tanks, pipelines, etc. Functional protective film provides properties including corrosion prevention, leakage prevention, reflective thermal insulation, and reduction of rain impact noise.

1.0.3

Where this specification does not provide requirements, current relevant national standards shall be implemented.

2 Terms

2.0.2

After construction using the fusion dry-bonding method, the later-stage bonding strength of the functional protective film is greater than the initial bonding strength, and the bond becomes tighter.

2.0.3

In addition to using cement-based adhesive materials for bonding, other adhesive materials that can firmly bond the functional protective film to the substrate surface and are compatible may also be used.

3 Basic Requirements

3.0.1

The importance of the protective engineering function considers the social, economic, and environmental impacts of leakage and corrosion. It mainly includes the following factors: (1) the impact of leakage and corrosion on instruments, equipment, materials, and other property inside the project; (2) the impact of leakage and corrosion on normal service conditions, structural durability, structural safety, etc.; and (3) repair cost and repair difficulty after leakage and corrosion. For roofs, exterior walls, and similar works, there are many climate factors affecting the service environment, including annual precipitation, extreme temperature, temperature difference, solar radiation, and wind load.

3.0.2

The metal surfaces specified in this article include uncoated metal panel surfaces and coated metal panel surfaces. In addition to metal surfaces, substrate surfaces suitable for the fusion dry-bonding method include stone surfaces, tile surfaces, etc.

3.0.3

For renovation projects of existing buildings, where existing service functions are not changed and conditions make implementation of current standards genuinely difficult, the requirements shall not be lower than the standards at the time of original construction. The original construction standards shall be determined according to the original design documents. Replacement and reinforcement materials shall be made of the same material as, or materials compatible with, the original metal structure. Insulating materials shall be used for physical separation between different metal materials to avoid electrochemical corrosion and similar problems.

3.0.4

For other structural layers, such as thermal insulation layers and waterproofing layers, the thermal insulation layer may be implemented in accordance with relevant provisions of the Code for Thermal Design of Civil Buildings, GB 50176; Design Standard for Energy Efficiency of Residential Buildings in Severe Cold and Cold Zones, JGJ 26; Design Standard for Energy Efficiency of Residential Buildings in Hot Summer and Warm Winter Zones, JGJ 75; Design Standard for Energy Efficiency of Residential Buildings in Hot Summer and Cold Winter Zones, JGJ 134; Design Standard for Energy Efficiency of Public Buildings, GB 50189; and other relevant standards. The waterproofing layer may be implemented in accordance with Technical Code for Roof Engineering, GB 50345; Code for Acceptance of Construction Quality of Roof, GB 50207; General Code for Waterproofing of Building and Municipal Engineering, GB 55030; and other relevant standards. Anti-corrosion design may refer to Standard for Anti-Corrosion Design of Industrial Buildings, GB/T 50046, and other relevant standards.

4 Materials

4.0.2

Engineering practice and testing have shown that functional protective film has good compatibility with Portland cement or ordinary Portland cement. Other cements shall be verified before use. The water-cement ratio is controlled to ensure the strength and fluidity of the cement-based adhesive material, improve efficiency in adhesive bonding construction, and ensure the construction quality of the functional protective film.

5 Design

5.1 General Requirements

5.1.1

Metal surface protection design for buildings and structures shall be carried out according to climate conditions, form, appearance, and other factors of the project location. Functional protective film is prevention-oriented and mainly provides leakage prevention, corrosion prevention, cooling, and noise reduction, without greatly increasing the load on the original structure, thereby ensuring structural safety.

5.1.2

The overlap width of functional protective film is one of the important parameters for ensuring the quality of overlap edges. It directly affects wind-uplift resistance and the protective effect at overlap parts. Based on construction practice, this specification sets the minimum overlap width at not less than 30 mm. Technical Specification for Single-Ply Membrane Roofing, JGJ/T 316, states that when the fully bonded method is used, the peel strength between the waterproof sheet and base layer shall not be less than 30 N/50 mm, equivalent to 0.6 N/mm. The peel strength between the functional protective film and the base layer is not less than 1.0 N/mm, satisfying wind-uplift resistance and protection requirements.

5.1.3

Whether an additional layer is provided shall be determined according to actual project conditions. Where working space is limited or the structure is complex, additional layer installation may be difficult, and wrinkles or hollow areas may form water-channeling leakage paths. In such cases, the additional layer may be omitted.

5.3 Detail Design for Metal Roof Protection

5.3.2

Laying the functional protective film to the eaves drip line is intended to prevent water backflow from corroding and rusting the metal panel surface at the lower edge of the eaves, and also to increase wind-uplift resistance.

5.3.3

Eaves gutters and valley gutters are the most concentrated drainage areas. Sheet laying in eaves gutters and valley gutters shall start from the gutter bottom, and the sheet shall be overlapped in the direction of water flow. When the eaves gutter or valley gutter is a concrete or masonry plastered surface, cement-based adhesive material may be used to bond the functional protective film to the base layer. A wet-laid waterproofing membrane with sheath-core structure, hot-pressed crosslinked polymer carrier may also be used for wet-lay construction. This membrane can achieve sealed full bonding with cement-based substrate surfaces, provides durable adhesion, and prevents leakage and water channeling; a membrane protective layer shall be provided.

6 Construction

6.1 General Requirements

6.1.6

Requirements for the base layer are key to ensuring construction quality of the functional protective film. Corresponding treatment shall be carried out for metal substrate surfaces under different conditions. When the metal substrate surface is damaged or corroded and cannot meet service function requirements, replacement or reinforcement treatment shall be carried out to ensure structural safety. When corrosion of the metal substrate surface does not affect structural safety, the functional protective film may be laid after derusting. Derusting may be carried out by grinding, applying rust-treatment materials, or other methods.

6.1.8

Before construction, hazard sources shall be identified according to project characteristics, construction processes, site environment, climate conditions, and similar factors. Risk assessment shall be carried out and risk-control measures shall be formulated. In particular, areas with higher risk shall be equipped with corresponding fall-prevention equipment and safety facilities. When working at height, safety protection shall be carried out according to the requirements of Technical Code for Safety of Working at Height of Building Construction, JGJ 80-2016. Construction personnel shall wear safety helmets, protective shoes, and necessary labor protection equipment, and shall use safety belts. Firefighting facilities shall be provided at the construction site, and management of ignition sources shall be strengthened.

6.1.11

Fall-prevention measures include wearing safety belts, erecting protective platforms, installing safety nets, strengthening safety education, paying attention to weather conditions, and providing emergency equipment.

6.1.12

Staggering short-side overlap joints of two adjacent sheets of functional protective film in the same layer avoids four-layer overlap of functional protective film and prevents adverse effects on joint quality. Staggering long-side overlap joints of upper and lower layers avoids overlap of defects in the overlap joints of the functional protective film.

6.1.13

Installing the functional protective film on the metal substrate surface before other processes is intended to ensure effective protection. Finished-product protection measures for the functional protective film shall be taken during other processes. For example, when the waterproofing layer of a single-ply membrane roof has failed, after the original waterproofing layer is removed and functional protective film construction is carried out, whether to restore the waterproofing layer shall be determined by the relevant responsible parties.

6.1.14

Damaged areas of functional protective film shall be repaired promptly. The damaged area shall be cleaned, and the damaged part shall be patched with functional protective film. The distance from the patch edge to the damaged-area edge shall not be less than 30 mm. Air shall be expelled and the patch pressed flat.

6.2 Metal Roof Protection Construction

6.2.1

Eaves gutters and valley gutters are areas where rainwater concentrates, while overlap joints of functional protective film are weak points. If functional protective film is laid perpendicular to the direction of eaves gutters and valley gutters, overlap joints will greatly increase, the overlap direction will be difficult to control, and the probability of seam opening and water scouring of the functional protective film will increase. Therefore, this specification provides that functional protective film in eaves gutters and valley gutters should be laid in the direction of water flow, minimizing overlap joints as much as possible.

6.3 Metal Wall Protection Construction

6.3.2

When functional protective film is laid horizontally, the overlap direction shall follow the direction of water flow to prevent water flow from scouring overlap parts and causing leakage. When laid vertically, the overlap direction shall follow the prevailing wind direction to reduce damage to functional protective film caused by wind pressure.

7 Acceptance

7.1 General Requirements

7.1.4

Metal roof protection works may be inspected by water spraying. This article specifies the duration of the water spraying test.

7.1.5

This article specifies that metal exterior wall protection works shall be inspected using the whole-wall water spraying inspection method rather than post-rain observation. This is mainly because, when post-rain observation is used for leakage inspection, the uncertainty of combined rain and wind often makes it difficult to meet leakage inspection requirements.

8 Operation and Maintenance

8.0.1

After construction of functional protective film is completed, designated personnel shall be arranged for regular inspection and maintenance during operation. Damage or leakage shall be repaired immediately once found.